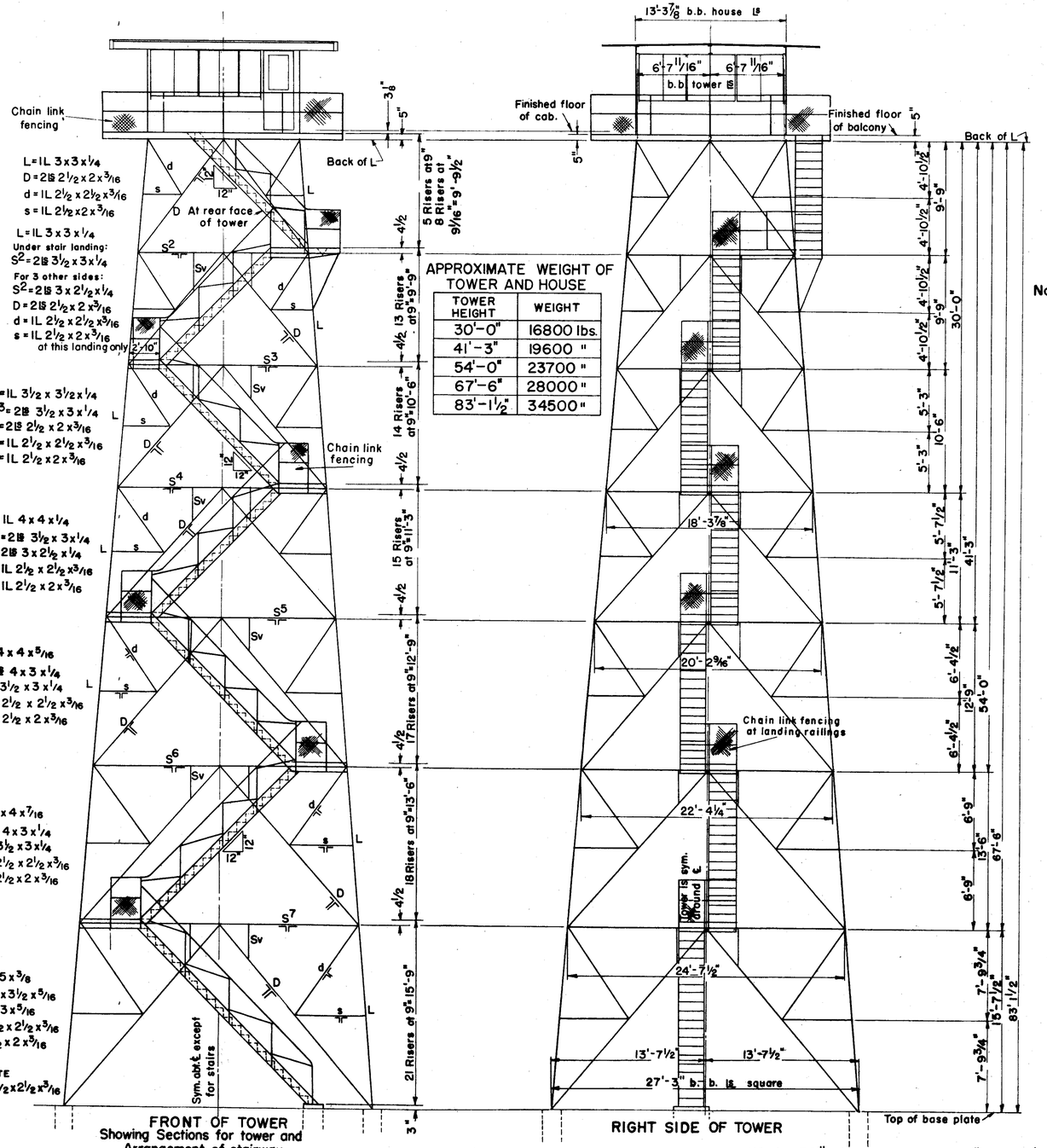


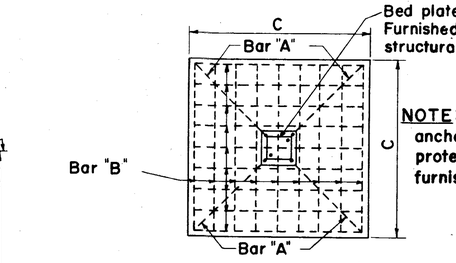
FRONT OF TOWER
Showing Sections for tower and Arrangement of stairway

RIGHT SIDE OF TOWER

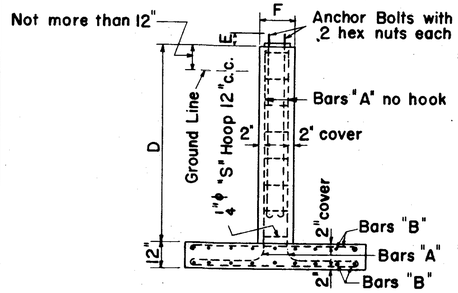
APPROXIMATE WEIGHT OF TOWER AND HOUSE

TOWER HEIGHT	WEIGHT
30'-0"	16800 lbs.
41'-3"	19600 "
54'-0"	23700 "
67'-6"	28000 "
83'-1 1/2"	34500 "

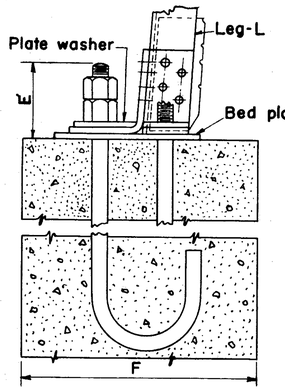
2 angle sections shall have long or short legs back to back as shown.
⌐⌐ sections to be field bolted with ring fillers at about 2 ft. intervals.
Bolts in tower members 3/4" ϕ except for connections as noted on details.
Gusset plates 5/16" for tower members.
Working lines as shown above are gage lines except as shown or noted.



NOTE: Concrete, reinforcing, anchor bolts, and lightning protection material to be furnished by the government.

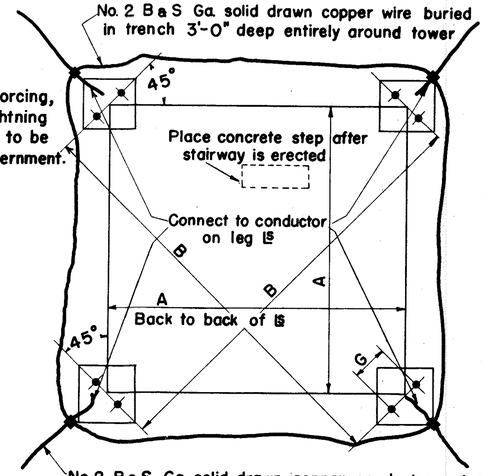


TYPICAL CONCRETE PIER



ANCHOR BOLTS IN CONCRETE

Note: Anchor bolts & reinforcing bars shall conform to F.S. QQ-B-71a, Grade 2. Anchor bolts & Hoop bars, Type A; Reinforcing bars, Type B.
Concrete Class A-1 1/2 aggregate. Mix 1-2-3 by volume 6 gal. of water per bag of cement.
Paint ends of anchor bolts with one coat of red lead before erecting tower and with a second coat of red lead and a final coat of aluminum after erection.
Reinforcing, anchor bolts, concrete materials and lightning protection items not furnished by steel fabricator.



FOUNDATION LAYOUT

SCHEDULE OF REINFORCING STEEL

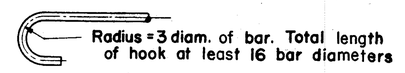
Tower Height	Bars "A" No.	Bars "A" Size	Bars "A" Length	Bars "B" No.	Bars "B" Size	Bars "B" Length	Hoops "S"	Wt. lbs.	Anchor Bolts No.	Anchor Bolts Diam.	Anchor Bolts Length	Anchor Bolts Wt.lbs
30'-0"	4	3/4" ϕ	7'-6"	16	1/2" ϕ	4'-0"	5	103	2	3/4"	3'-1"	13
41'-3"	4	7/8" ϕ	8'-3"	20	1/2" ϕ	4'-6"	5	145	2	3/4"	3'-1"	13
54'-0"	4	1" ϕ	9'-3"	24	1/2" ϕ	5'-0"	6	200	2	7/8"	3'-7"	21
67'-6"	4	1" ϕ	10'-0"	28	1/2" ϕ	5'-6"	6	263	2	1"	4'-2"	30
83'-1 1/2"	4	1 1/8" ϕ	10'-9"	32	1/2" ϕ	6'-0"	7	298	2	1"	4'-2"	30

Note: Length given is finished length unless noted. Bars A Total length, no hook Bar size No.9 and No.10 may be substituted for the 1" ϕ and 1 1/8" ϕ bars "A" listed. Quantities listed are for one pier only.

PIER DIMENSIONS

Tower Height	A	B	C	D	E	F	G	Cu.ft. Conc.
30'-0"	18'-3 7/8"	25'-10 15/16"	4'-6"	5'-0"	3 3/4"	1'-4"	4 15/16"	29
41'-3"	20'-2 9/16"	28'-7 1/16"	5'-0"	5'-6"	4"	1'-4"	5 1/16"	35
54'-0"	22'-4 1/4"	31'-7 3/8"	5'-6"	6'-0"	4 1/2"	1'-4"	5 1/16"	41
67'-6"	24'-7 1/2"	34'-9 7/8"	6'-0"	6'-6"	5"	1'-4"	5 1/16"	48
83'-1 1/2"	27'-3"	38'-6 7/16"	6'-6"	7'-0"	5"	1'-4"	7 1/16"	55

Concrete listed is for one pier only



TYPICAL HOOK

DESIGN DATA

Loading

Wind - 30#/□' on house and 30#/□' on 1 1/2 times exposed area of tower and stairs. Tower designed for wind blowing diagonally by increasing wind load stresses in tower members by 41.4%.
Live load- House roof 65#/□', house floor 50#/□', catwalk 50#/□', stairs 100#/□' on horizontal projection.

Allowable Stresses

Tension 20000#/□" on net section.
Compression
Columns, Main Diagonals, & Struts ($\frac{l}{r}$ not to exceed 120) $17000 - 0.485(\frac{l}{r})^2$
Secondary Diagonals & Struts ($\frac{l}{r}$ not to exceed 200) $18000 \Big/ 1 + \frac{l^2}{18000r^2}$

Rivets & Ribbed Bolts
Shear 15000#/□"
Bearing 40000#/□" DS, 32000#/□" SS
Unfinished Bolts
Shear 10000#/□"
Bearing 25000#/□" DS, 20000#/□" SS

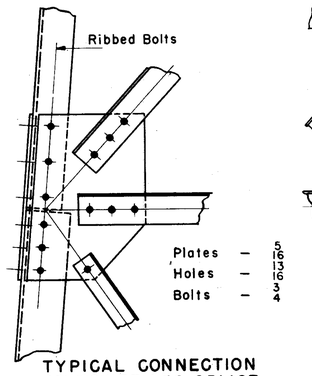
Revised Jan. 1951, Jan. 1954 954-[illegible]

U. S. DEPARTMENT OF AGRICULTURE
FOREST SERVICE

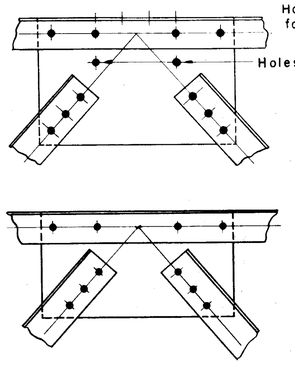
STANDARD STEEL LOOKOUT TOWERS
30', 41'-3", 54', 67'-6" & 83'-1 1/2" HIGH
WITH LIVING QUARTERS
ELEVATION, MEMBERS and FOUNDATION

DESIGNED REGION 7 DRAWN REGION 7 TRACED D.B.S.
SCALE CHECKED J.H.
APPROVED A. P. Stern Chief Engineer DATE 2-23-51
SHEET 1 OF 6

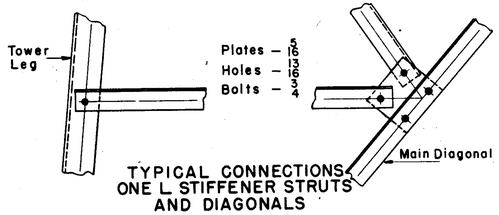
C-L-101



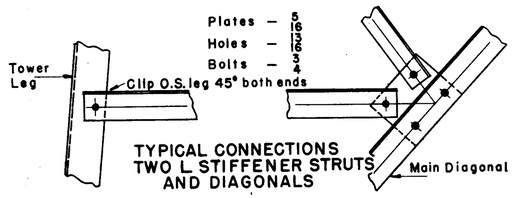
TYPICAL CONNECTION AT TOWER LEG SPLICE



TYPICAL CENTER CONNECTIONS

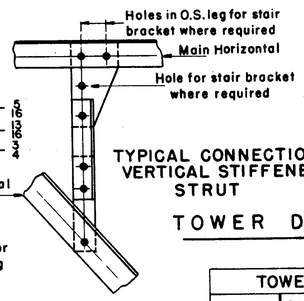


TYPICAL CONNECTIONS ONE L STIFFENER STRUTS AND DIAGONALS

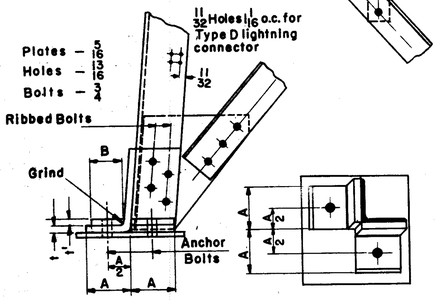


TYPICAL CONNECTIONS TWO L STIFFENER STRUTS AND DIAGONALS

All 2 angle sections to be field bolted at about 2 ft. intervals. Provide ring fillers.

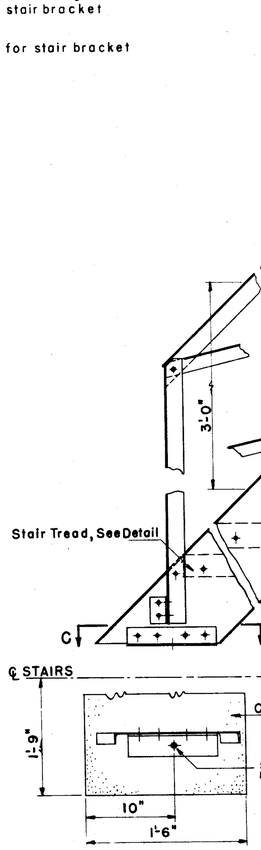


TYPICAL CONNECTIONS VERTICAL STIFFENER STRUT



TYPICAL BASE DETAIL

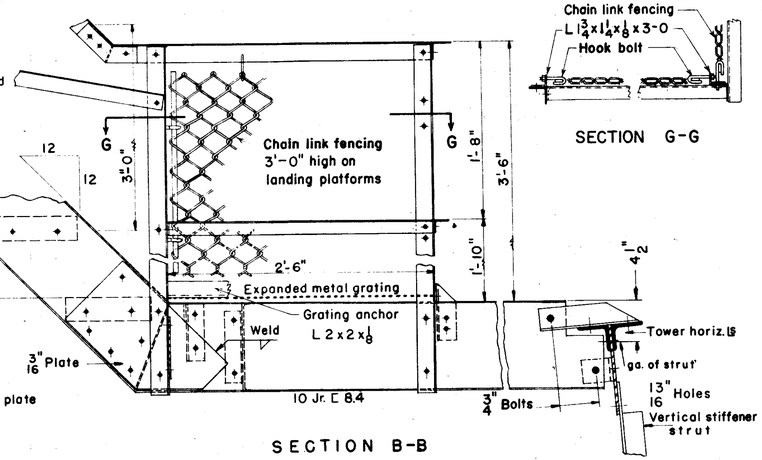
TOWER DETAILS



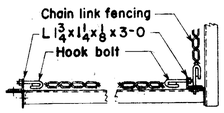
SECTION A-A

STAIR DETAILS

Stringers — 10 Jr C 8.4
Rail Posts — L 2x2x1/8 unless noted
Top Rail — L 2x2x1/8
Lower Rail — L 1 1/2x1 1/2x1/8
Rail Struts — L 1 1/2x1 1/2x1/8
Gusset Plates — 7/16
Bolts — 3/8, Holes 7/16 unless noted
Landings — Expanded metal grating
Treads — See Detail
Chain link fencing to be provided at landing platforms

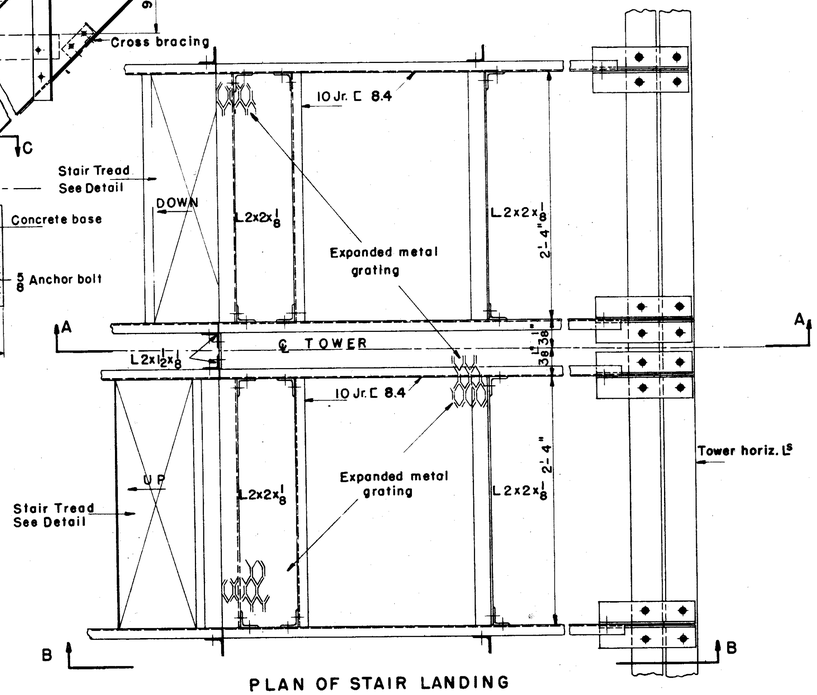


SECTION B-B

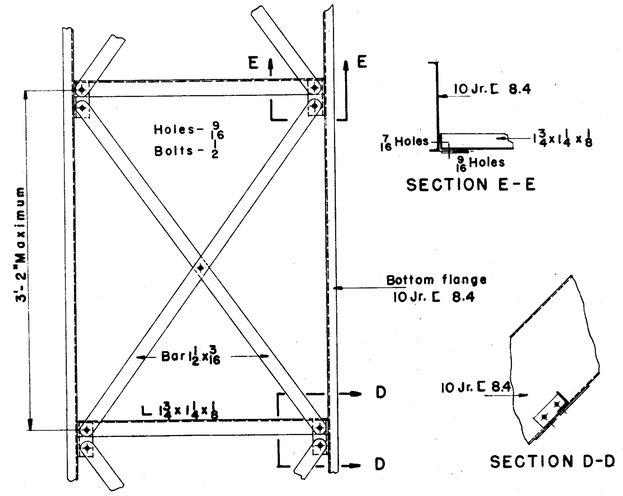


SECTION G-G

SECTION C-C

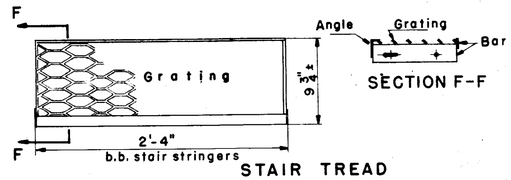


PLAN OF STAIR LANDING



STAIR STRINGER CROSS BRACING

SECTION E-E



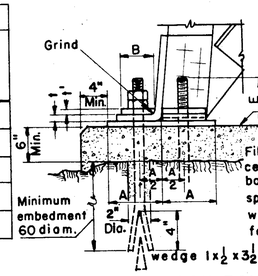
SECTION F-F

STAIR TREAD

STAIR DETAILS

TOWER POST BASE MATERIALS

TOWER HEIGHT	Base L A	Base L t	Bolts per Tower Leg	Plate Washer B	Plate Washer t'	Plate Washer A	Base Plate
30'-0"	3 1/2"	1/2"	6	2 3/4"	1/2"	3 1/2"	10"x 1/2" x 10"
41'-3"	4"	5/8"	6	3"	1/2"	4"	10"x 1/2" x 10"
54'-0"	4"	5/8"	6	3	5/8"	4"	10"x 1/2" x 10"
67'-6"	4"	3/4"	6	3"	5/8"	4"	10"x 1/2" x10"
83'-1 1/2"	5"	3/4"	8	3 3/4"	5/8"	5"	12"x 1/2" x12"



ROCK FOUNDATION

ANCHOR BOLTS IN ROCK FOUNDATION

TOWER HEIGHT	Bolt	E	Weight Lbs. 8 Bolts
30'-0"	3/4" x 4'-6"	3 3/4"	63
41'-3"	3/4" x 4'-6"	4"	63
54'-0"	7/8" x 5'-2"	4 1/2"	95
67'-6"	1" x 6'-0"	5"	143
83'-1 1/2"	1" x 6'-0"	5"	143

Revised Dec. 1954 APD
Revised Jan. 1954 APD

U. S. DEPARTMENT OF AGRICULTURE
FOREST SERVICE

STANDARD STEEL LOOKOUT TOWERS
30', 41'-3", 54', 67'-6" & 83'-1 1/2" HIGH
WITH LIVING QUARTERS

TOWER & STAIR DETAILS

DESIGNED R-7 DRAWN R-7 TRACED N.H.O.
SCALE CHECKED
APPROVED A P Dean Chief Engineer DATE 3-3-51
SHEET 2 OF 6

C — L-102

PLAN OF LANDING AT 9'-9

RAILING DETAILS AT CORNERS OF PLATFORM

RAILING DETAILS AT CORNERS OF LANDING

TRAP DOORS

DETAIL A

SECTION L-L

PLAN AT CAB LEVEL

SECTION B-B

SECTION A-A

ELEVATION J-J

SECTION K-K, SECTION M-M Similar

SECTION C-C

SECTION D-D

SECTION H-H

SECTION E-E

SECTION G-G

SECTION F-F

TYPICAL ANCHORAGE FOR PLATFORM GRATING

NOTE-

All holes to be $\frac{13}{16}$" in tower details and $\frac{7}{16}$" in platform, landing, stair, and railing details unless otherwise noted.

Platform and landing decking shall be anchored to structural framing by means of hook bolts and clips or by 2 x 2 x $\frac{1}{8}$ angles as shown on details.

Revised Dec. 1954 APD
Revised Jan. 1954

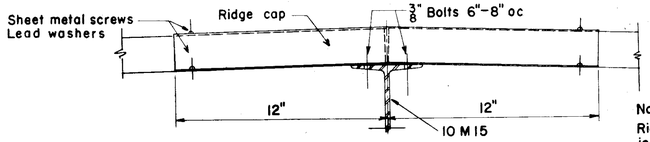
U. S. DEPARTMENT OF AGRICULTURE
FOREST SERVICE

STANDARD STEEL LOOKOUT TOWERS 30', 41'-3", 54', 67'-6", & 83'-1½" HIGH WITH LIVING QUARTERS

PLATFORM DETAILS

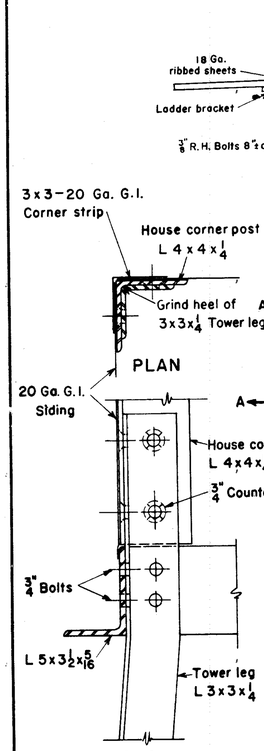
DESIGNED R-7 DRAWN R-7 & W.L.D. TRACED W.L.D.
SCALE CHECKED
APPROVED Chief Engineer DATE 5-5-51
SHEET 3 OF 6

C-L-103

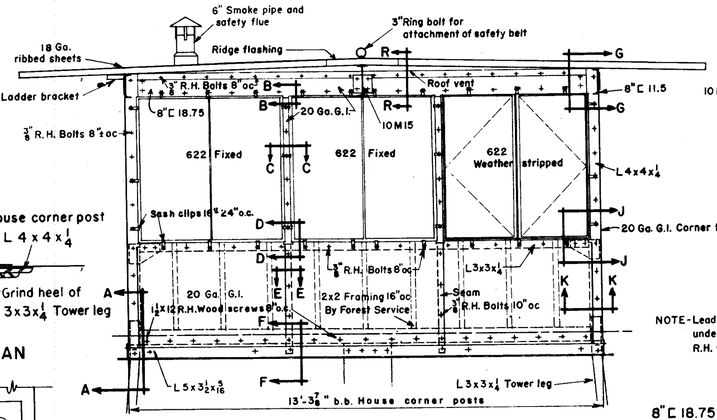


Note - Ridge cap to be set in mastic

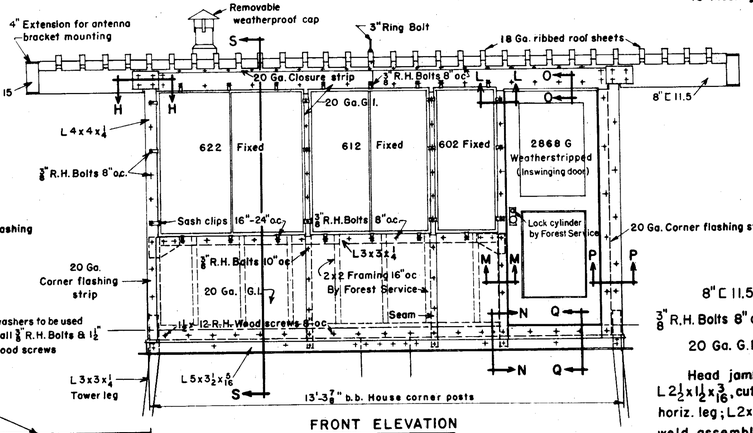
DETAIL OF RIDGE CAP



A-A
CORNER POST SPLICE TO TOWER LEG

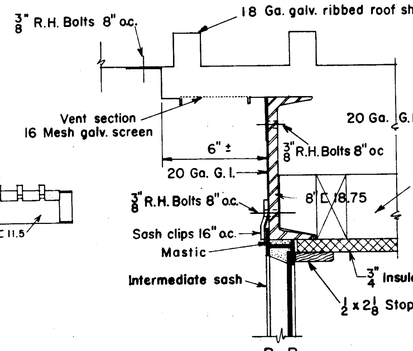


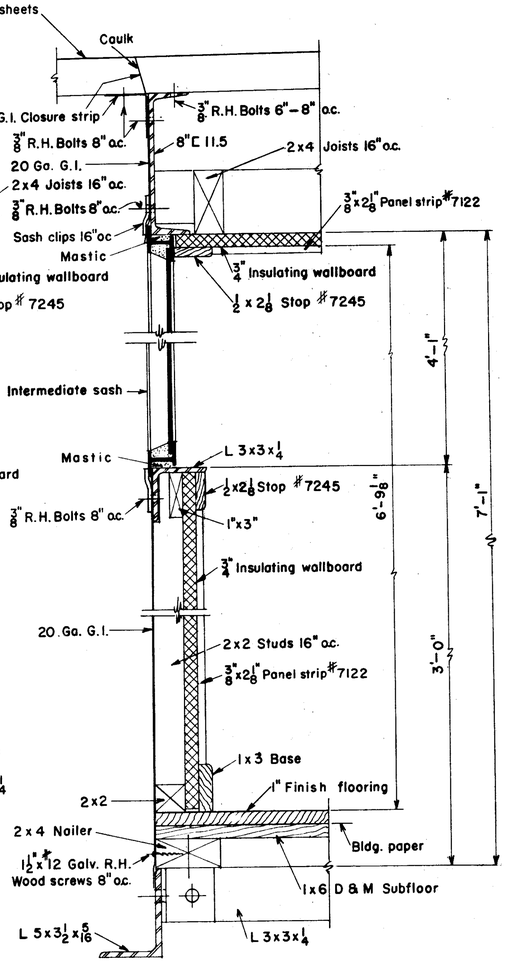
LEFT SIDE ELEVATION
RIGHT SIDE & REAR SIMILAR
NOTE: All sash in rear elevation to be fixed sash.



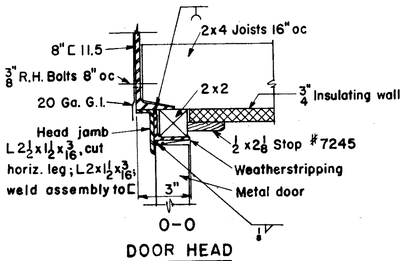
NOTE - Lead washers to be used under all 3/8" R.H. Bolts & 1 1/2" R.H. wood screws

FRONT ELEVATION

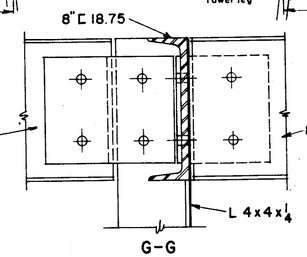




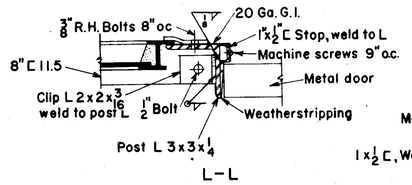
TYPICAL WALL SECTION
See Section S-S



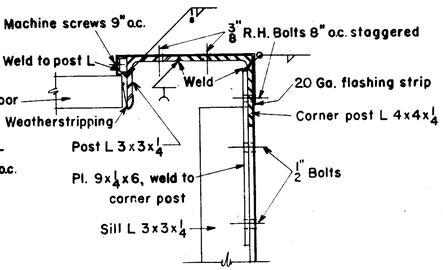
DOOR HEAD



G-G

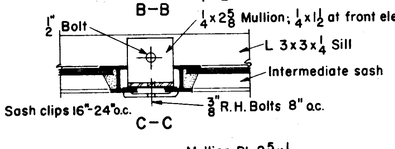


L-L

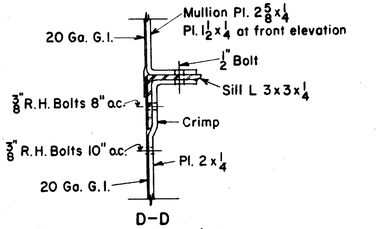


P-P

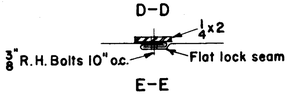
B-B



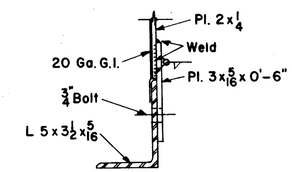
C-C



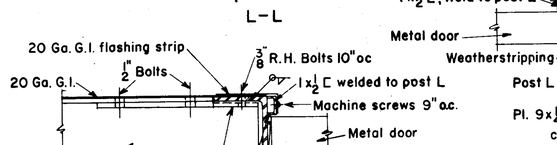
D-D



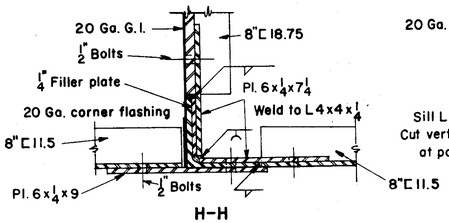
E-E



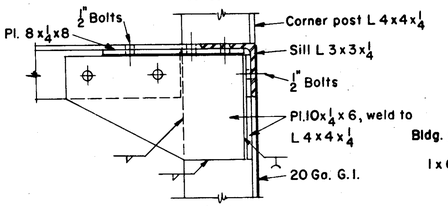
F-F
FRAMING DETAILS AT WINDOW MULLIONS



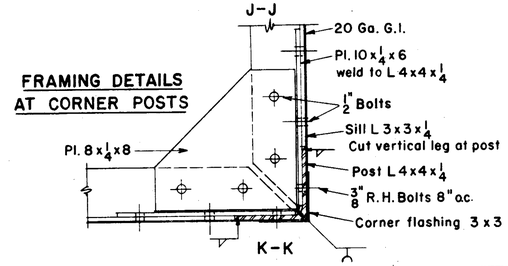
M-M



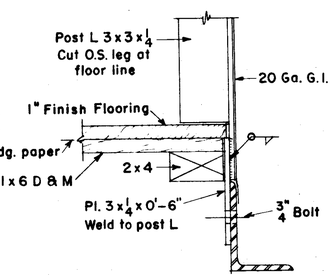
H-H



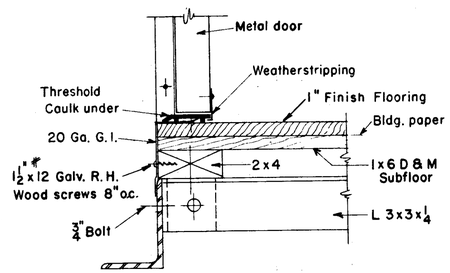
J-J



K-K
FRAMING DETAILS AT CORNER POSTS



N-N
FRAMING AT DOOR POST



Q-Q
DETAIL AT DOOR

NOTE
See Sheet 5 for floor and roof framing details

NOTES TO FOREST SERVICE
1-House framing lumber, flooring, wallboard and interior trim not furnished by steel fabricator.
2-Rock wool or glass fiber insulation may be installed in walls and ceiling space, if desired. Material not furnished by steel fabricator.
3-Interior wall and ceiling surfaces of house to be field painted with light green gloss paint. Paint not furnished by steel fabricator.

NOTES
Unless otherwise noted, connections shall be as follows:
Structural framing - 1/2" Bolts with lock nuts; 9/16" holes
Sheet metal - 3/8" R.H. Bolts with lock nuts; 1/2" Holes
Lead washers required at contact surface between head of fasteners and sheet metal, except for structural fasteners

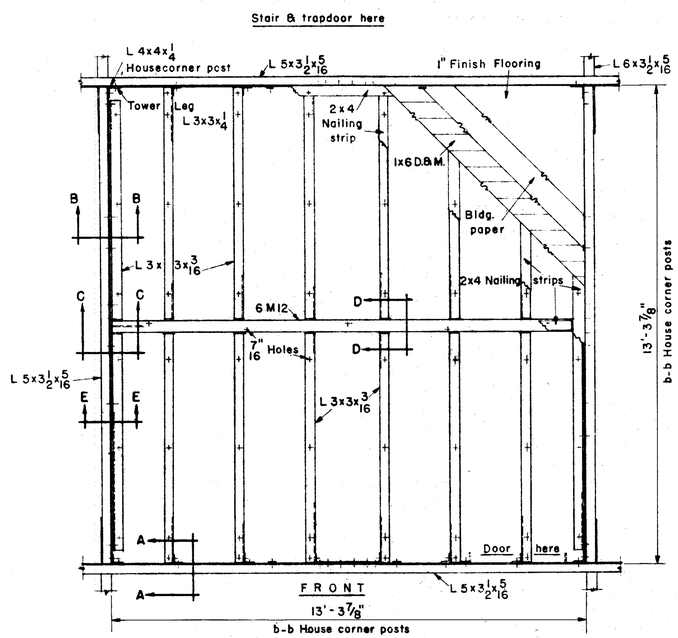
U. S. DEPARTMENT OF AGRICULTURE
FOREST SERVICE

STANDARD STEEL LOOKOUT TOWERS 30', 41'-3", 54', 67'-6", & 83'-11 1/2" HIGH WITH LIVING QUARTERS

HOUSE DETAILS

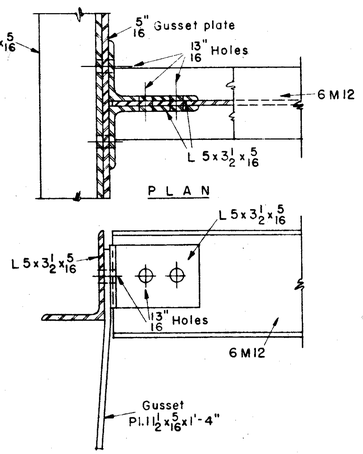
DESIGNED NHQ DRAWN NHQ-WLD TRACED
SCALE CHECKED
APPROVED C. P. Dean Chief Engineer DATE Dec. 1954
SHEET 4 OF 6

C-L-104

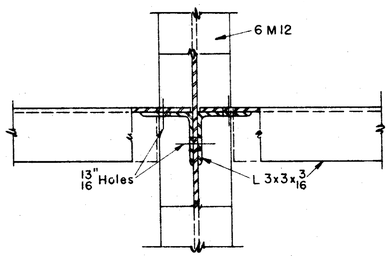


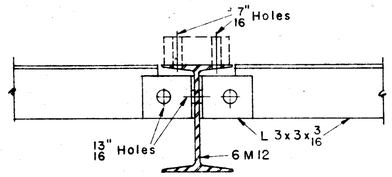
FLOOR FRAMING

All holes in this framing to be 13/16" unless otherwise noted.
2x4 Nailers and 1x6 D&M subfloor to be pressure treated with pentachlorophenol.
Nailers and flooring not furnished by steel fabricator. Bolts for 2x4 nailer to be furnished.

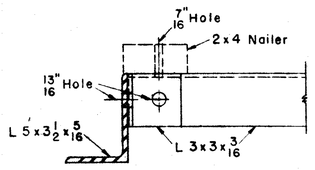


C - C

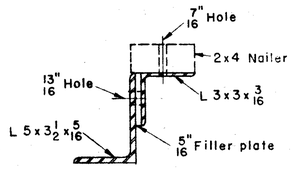




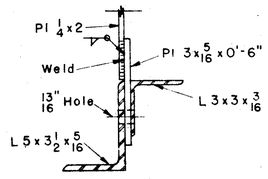
D - D



A - A

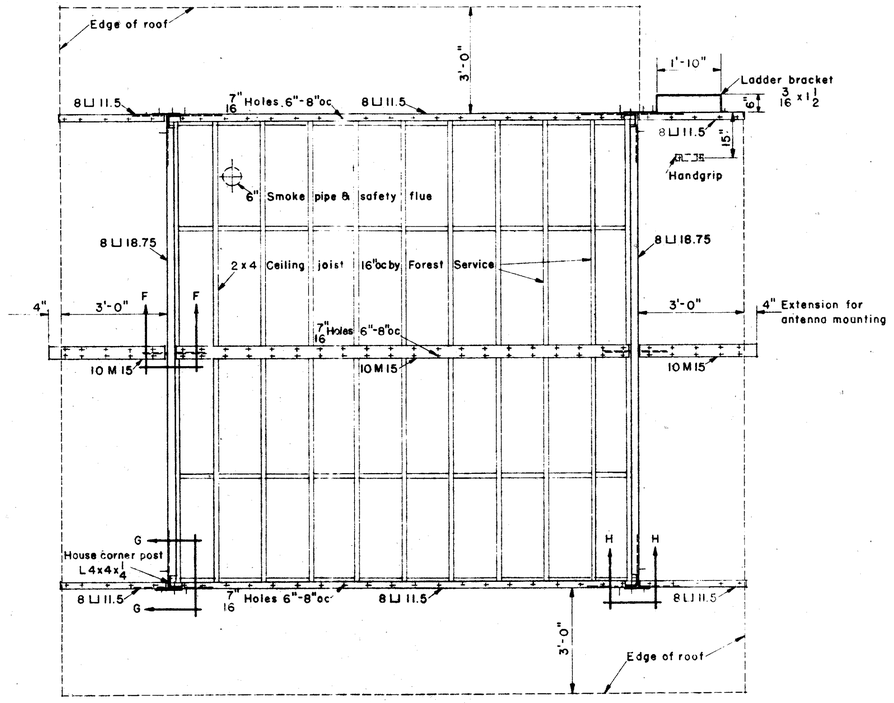


B - B



E - E

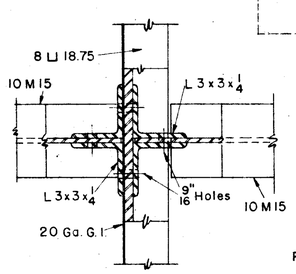
FLOOR FRAMING DETAILS



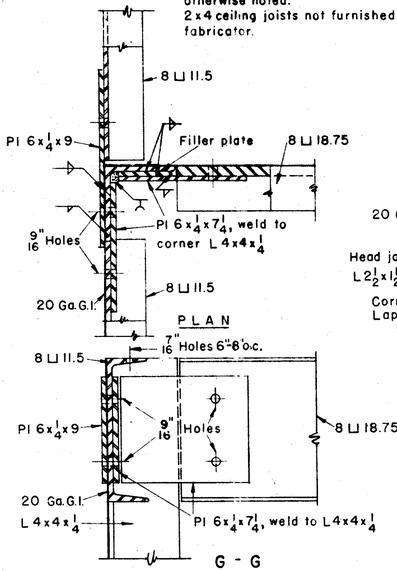
FRONT

ROOF FRAMING

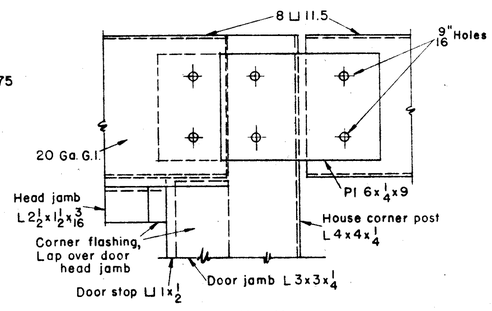
All holes in this framing to be 9/16" unless otherwise noted.
2x4 ceiling joists not furnished by steel fabricator.



F - F



G - G



H - H

ROOF FRAMING DETAILS

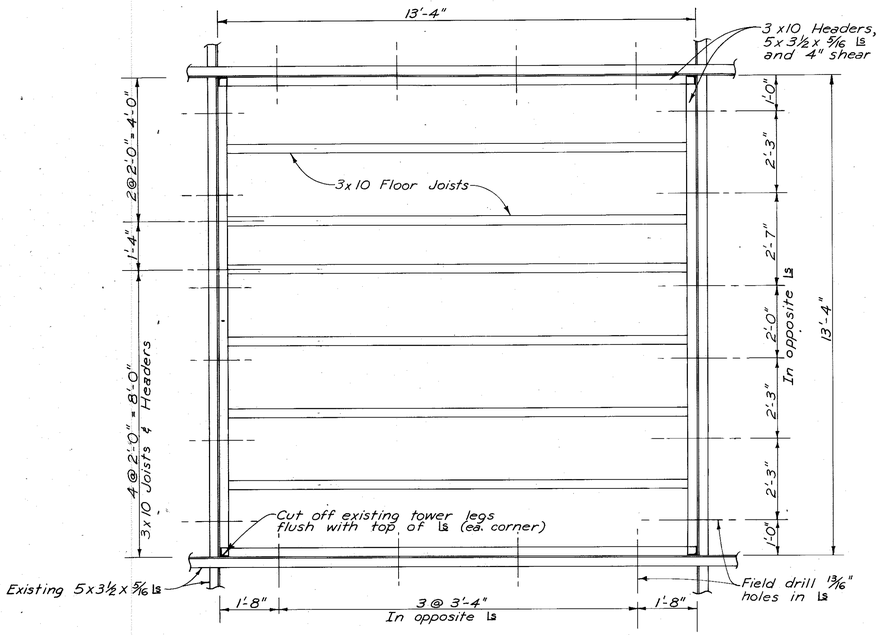
U. S. DEPARTMENT OF AGRICULTURE
FOREST SERVICE

STANDARD STEEL LOOKOUT TOWERS
30', 41'-3", 54', 67'-6", & 83'-1 1/2" HIGH
WITH LIVING QUARTERS
HOUSE DETAILS

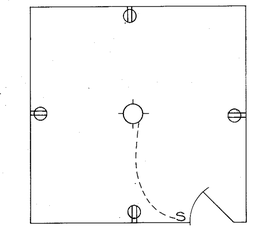
DESIGNED NHO DRAWN NHO - RR TRACED
SCALE CHECKED NHO
APPROVED A. Ortega Chief Engineer DATE Dec. 1954
SHEET 5 OF 6

C - L-105

ALBUQUERQUE BLUE PRINT 200 9-73 MC1394•

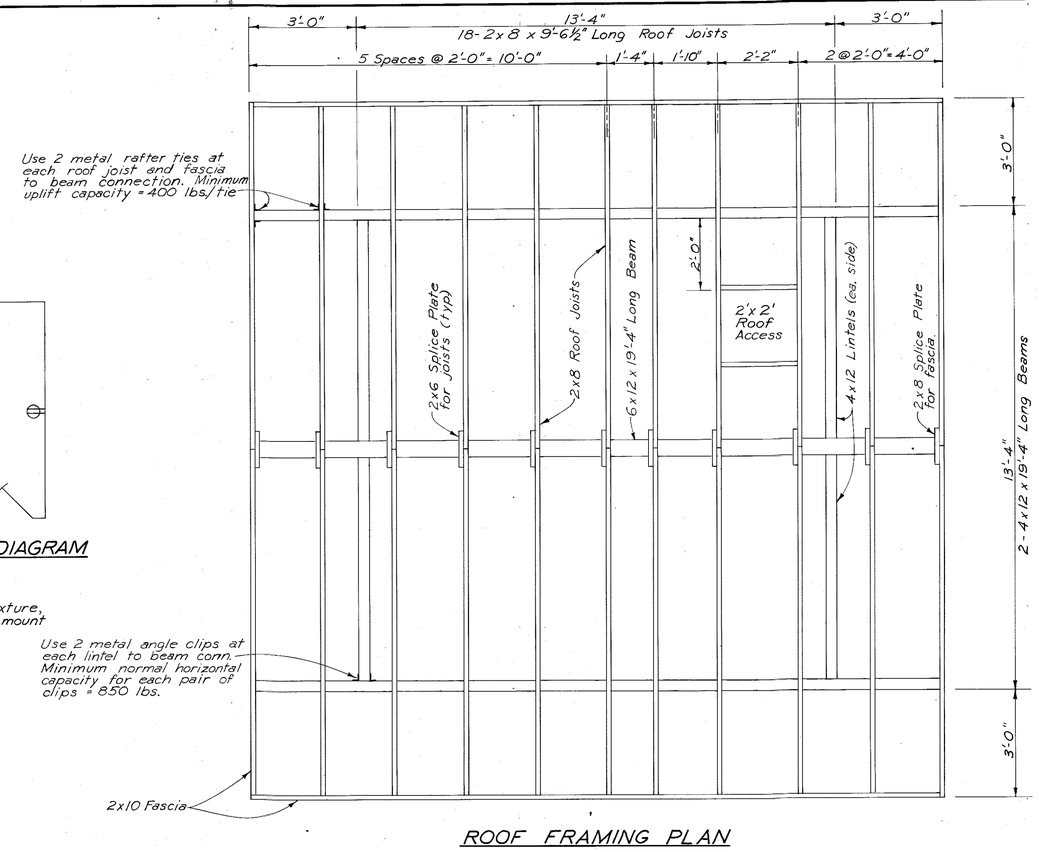


FLOOR JOIST FRAMING PLAN

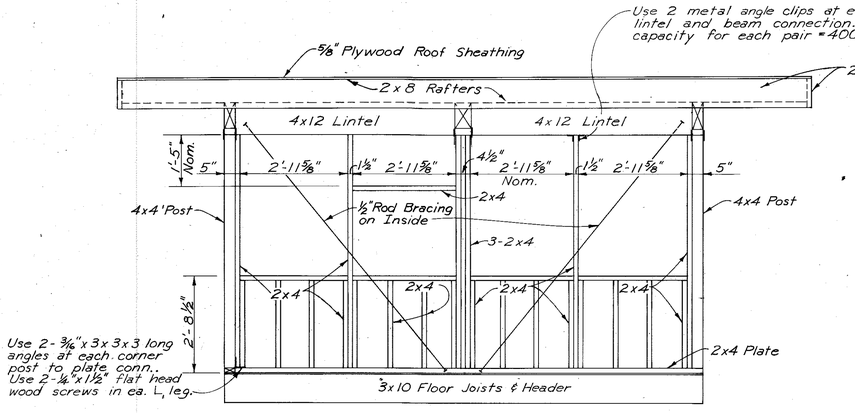


ELECTRIC WIRING DIAGRAM

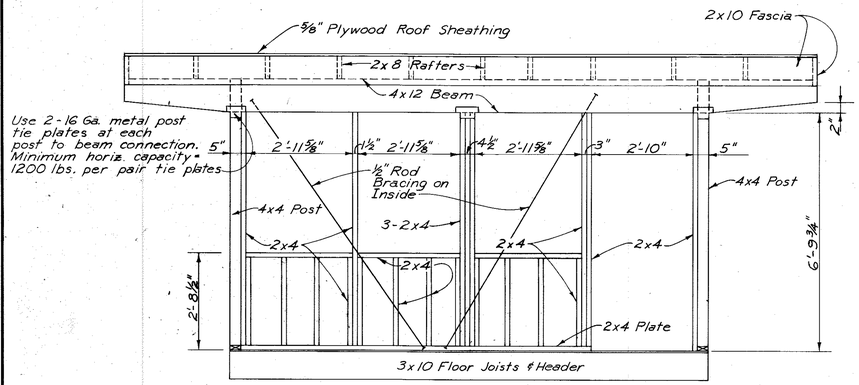
- Double Outlet
- S Single Pole Switch
- Single bulb light fixture, ceramic base, ceiling mount



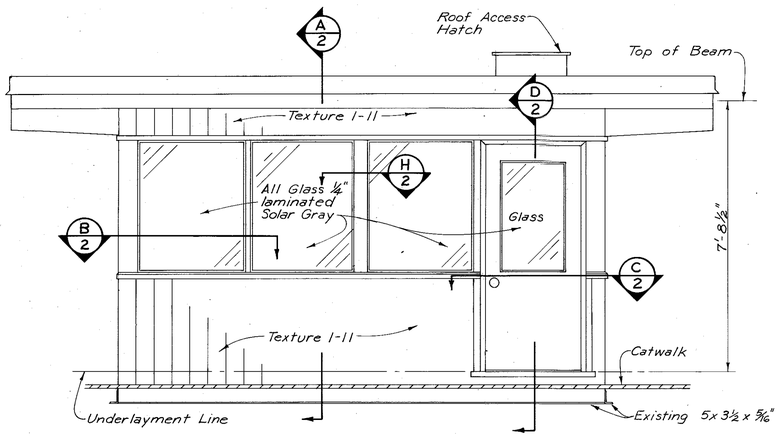
ROOF FRAMING PLAN



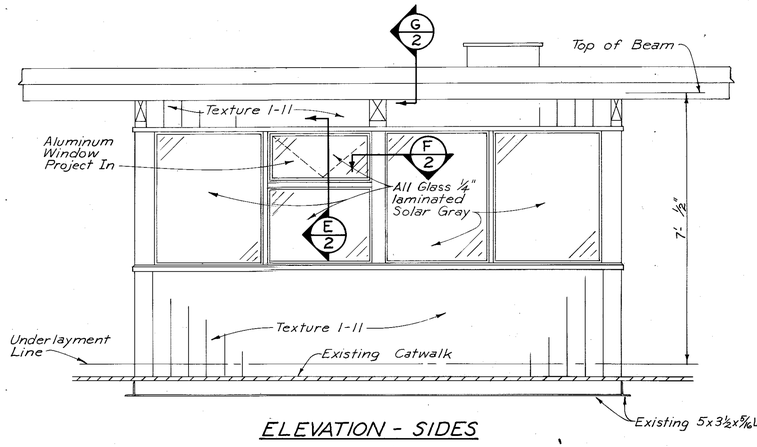
ROUGH FRAMING - SIDES



ROUGH FRAMING - FRONT



ELEVATION - FRONT
(Door Faces East)



ELEVATION - SIDES

NOTES:

Loads: Cab floor and catwalk - 50 lbs./sq. ft., roof - 65 lbs./sq. ft., and wind - 40 lbs./sq. ft.

Glass: All glass ¼" thick laminated safety glass tinted solar gray.

Lumber: All ponderosa pine, No. 1 grade for joists, beams, and posts. Construction grade for framing members.

Plywood: C-D Interior grade for sub-floor, underlayment, and exterior wall sheating. A-C Interior grade for interior walls. A-C Exterior grade for outside surfaces, shutters, and roof sheating.

U. S. DEPARTMENT OF AGRICULTURE
FOREST SERVICE
REGION THREE

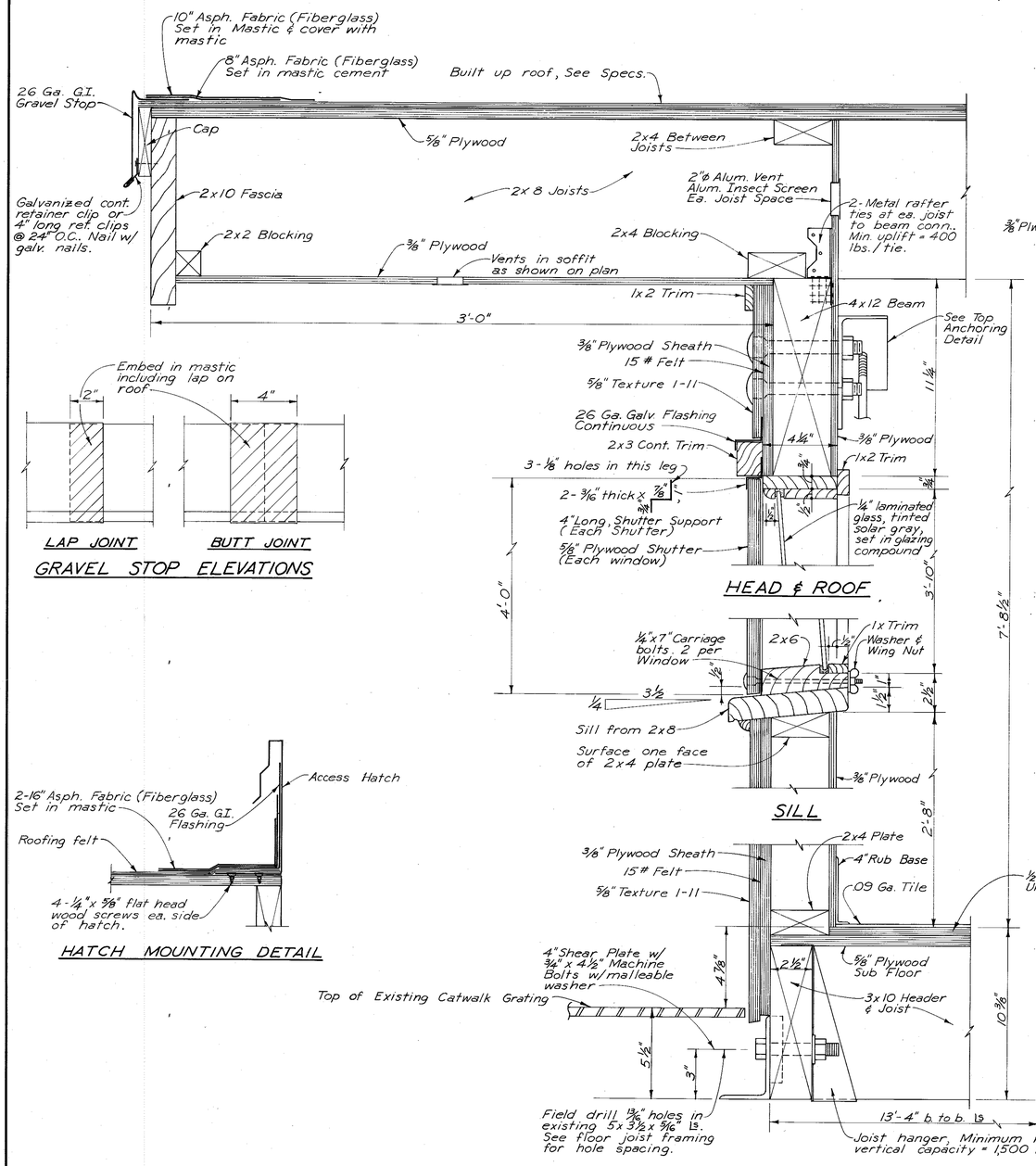
COCONINO NATIONAL FOREST — ARIZONA

MT. ELDEN LOOKOUT CAB

DESIGNED BDE DRAWN BGH TRACED
SCALE NO SCALE CHECKED
APPROVED DATE 5/12/78

Sheet 1 of 2

ALBUQUERQUE BLUE PRINT 200 9-73 MC1394



U. S. DEPARTMENT OF AGRICULTURE
FOREST SERVICE
REGION THREE

COCONINO NATIONAL FOREST — ARIZONA

MT. ELDEN LOOKOUT CAB

DESIGNED BDE DRAWN BGH TRACED
SCALE NO. SCALE CHECKED
APPROVED DATE 5/12/78